

MAT 524: Functions of a Real Variable II

Homework VIII

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[[Problem One]]:

Problem Statement

Let $f(x) = \frac{1}{2} - x$ on $[0, 1]$, viewed as an element of $L^2([0, 1])$. Show that $\hat{f}(0) = 0$ and $\hat{f}(n) = \frac{1}{2\pi in}$ if $n \neq 0$.

Apply Parseval's identity to compute $\zeta(2)$:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Solution

Note that the set $\mathcal{B} = \{e^{2\pi inx} \mid n \in \mathbb{Z}\}$ is an orthonormal basis for $L^2([0, 1])$. Using this set, we have

$$\hat{f}(0) = \langle f(x), e^0 \rangle \tag{1}$$

$$= \int_0^1 \left(\frac{1}{2} - x \right) dx \tag{2}$$

$$= \left[\frac{1}{2}x - \frac{x^2}{2} \right]_0^1 \tag{3}$$

$$= \frac{1}{2} - \frac{1}{2} \tag{4}$$

$$= 0. \tag{5}$$

For $n \neq 0$, we have

$$\hat{f}(n) = \langle f(x), e^{2\pi inx} \rangle \tag{6}$$

$$= \int_0^1 \left(\frac{1}{2} - x \right) \overline{e^{2\pi inx}} dx \tag{7}$$

$$= \int_0^1 \left(\frac{1}{2} - x \right) e^{-2\pi inx} dx \tag{8}$$

$$= \int_0^1 \frac{1}{2} e^{-2\pi inx} dx - \int_0^1 x e^{-2\pi inx} dx. \tag{9}$$

For the first integral in Equation (9), we have

$$\int_0^1 \frac{1}{2} e^{-2\pi i n x} dx = \frac{1}{-4\pi i n} [e^{-2\pi i n} - 1] \quad (10)$$

$$= 0. \quad (11)$$

To compute the second integral from Equation (9), we use integration by parts:

$$-\int_0^1 x e^{-2\pi i n x} dx = -\left[\frac{-x}{2\pi i n} e^{-2\pi i n x} \right]_0^1 - \int_0^1 \frac{1}{2\pi i n} e^{-2\pi i n x} dx \quad (12)$$

$$= \frac{1}{2\pi i n} - \left[\frac{1}{-4\pi^2 i^2 n^2} e^{-2\pi i n x} \right]_0^1 \quad (13)$$

$$= \frac{1}{2\pi i n}. \quad (14)$$

Thus, $\hat{f}(n) = \frac{1}{2\pi i n}$.

Now, since $\mathcal{B}$ is an orthonormal basis for $L^2([0, 1])$, Bessel's inequality becomes Parseval's identity. So

$$\sum_{n=-\infty}^{\infty} |\hat{f}(n)|^2 = 2 \sum_{n=1}^{\infty} \left| \frac{1}{2\pi i n} \right|^2 \quad (15)$$

$$= \sum_{n=1}^{\infty} \frac{1}{2\pi^2 n^2} \quad (16)$$

$$= \|f(x)\|^2 \quad (17)$$

$$= \int_0^1 \left| \frac{1}{2} - x \right|^2 dx \quad (18)$$

$$= \int_0^{0.5} (0.5 - x)^2 dx + \int_{0.5}^1 (x - 0.5)^2 dx \quad (19)$$

$$= \left[0.25x - \frac{x^2}{2} + \frac{x^3}{3} \right]_0^{0.5} + \left[\frac{x^3}{3} - \frac{x^2}{2} + 0.25x \right]_{0.5}^1 \quad (20)$$

$$= \frac{1}{12}. \quad (21)$$

Therefore,

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

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